

COMPUTER VISION & AI ALIGNMENT

Framework for Hand Related Video Annotation

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ABSTRACT

This paper defines a framework for creating high-quality egocentric video datasets. By applying a "Hand-Object Interaction" mental model, we can reduce hallucination risks and ensure clean segmentation for AI model training.

1

ANNOTATION FOUNDATION

The Core Mental Model: Goal-Oriented Interaction

The foundation of high-fidelity annotation is the "**One Segment = One Goal**" rule.

- **Action Identification**

A label is only required when hands are interacting with an object toward a specific goal.

- **No Action (NA)**

If hands have no contact with a task-relevant object, it is categorized as No Action.

- **Boundary Precision**

Segments must start exactly when engagement begins and end when the hands disengage or the goal changes.

2

DATASET PURITY

Decision Logic: Coarse vs. Dense

To maintain dataset purity, a segment must be either **Dense** or **Coarse**, never both.

- **Coarse Labels**

Preferred when a single goal verb (e.g., "wash dish") accurately covers a repetitive task without increasing hallucination risk.

- **Dense Labels**

Required when multiple distinct movements must be explicitly correct to describe the task (e.g., "insert wire, tighten screw").

3

CLEAN TEXT INPUT

Linguistic Standardization for Machine Learning

AI models require "clean" text input. This framework applies strict grammatical constraints to remove linguistic noise.

- **Structural Caps**

Labels should remain under 20 words and focus on 1-2 atomic actions.

- **Article Omission**

The use of "the," "a," or "an" is strictly forbidden.

- **Imperative Voice**

All labels must use direct action verbs attached to objects.

- **Objective Vocabulary**

Vague terms such as "inspect," "check," or "examine" are excluded to prevent subjective bias.

4

MASTERY CRITERIA

Quality Benchmarks & Standards

Mastery of this framework is defined by **zero tolerance** for "hallucinated" actions — inferring things that cannot be defended by on-screen evidence. Boundaries must reflect the *intent* of the actor rather than just the physical motion.